

Roll No.

TPE-601

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

MID SEMESTER EXAMINATION, April/May, 2022

RESERVOIR MODELING AND SIMULATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain *five* physical properties of rock. 10 Marks (CO1)

OR

- (b) Name the different type of rocks. Explain Sedimentary rock. How many type of sedimentary rocks are available on the earth ?

10 Marks (CO1)

2. (a) Draw a cross section of a petroleum system and explain its all important components. 10 Marks (CO1)

OR

- (b) Draw a diagram of a Petroleum system showing Generation, Migration and Trapping of hydrocarbons. 10 Marks (CO1)

P. T. O.

(2)

3. (a) Explain Reservoir Modeling , Reservoir Engineering and various steps of Reservoir Engineering. 10 Marks (CO2)

OR

- (b) Explain natural and artificial recovery mechanisms in details. 10 Marks (CO2)

4. (a) What is traps ? Illustrate different type of traps by drawing a neat figure. 10 Marks (CO2)

OR

- (b) Explain Reservoir pressure, Rock texture, Porosity, Permeability and Saturation. 10 Marks (CO2)

5. (a) Explain Darcy equation. How is it used to measure permeability for flowing fluid and gas system ? 10 Marks (CO2)

OR

- (b) Define surface and interfacial tension with a diagram. 10 Marks (CO2)

Roll No.

TPE-602

B. TECH. (PETROLEUM ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

DATA SCIENCE

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) DataBase Name : University 10 Marks (CO2)

TableName :

- (i) Student (columns: StudentID, name, batch, course, department)
- (ii) Department (Columns: DepartmentID, Name, HOD)

Write the query for the following assuming above details :

- (i) Print all the details of the students who are perusing "Petroleum Engineering".
- (ii) Print the studentid, name and name of their HOD.
- (iii) Print the departmentid of the 'petroleum' department

P. T. O.

OR

(b) Database Name : ABC Software Pvt. Ltd.

10 Marks (CO2)

Table Name :

(1) Employee (Column name : ID, Name, Project, Department, Salary, Manager)

Write the query for the following assuming above details :

(i) Print all the details of the employee whose salary is in the range 30000 and 100000

(ii) Print the name of manager whose employees are working on 'XYZ' project.

(iii) Print the employee id of the employee whose is 'RADHESHYAM'.

2. (a) Write short notes on the following :

10 Marks (CO1)

(i) Data Science

(ii) Machine Learning

(iii) Data Analytics

OR

(b) Describe your understanding on the Analytics. Also, discuss the analytics methodology you follow for solving the analytical problem.

10 Marks (CO1)

3. (a) Companies use data analytics to increase the revenue of the company. Explain how companies can get the benefits from data analytics.

10 Marks (CO1)

OR

(b) What are the 7 V's of Bigdata ? Explain each.

10 Marks (CO1)

4. (a) Write the syntax for the following with example : 10 Marks (CO2)

- (i) Left outer join
- (ii) Inner join
- (iii) Having clause
- (iv) Right outer join

OR

(b) Write the SQL queries for the following : 10 Marks (CO2)

- (i) Create the database with name as midterm
- (ii) Create table midtermschedule with columns as follows :
 - (a) SubjectCode—Varchar
 - (b) SubjectName—Varchar
 - (c) Date—Varchar
 - (d) Time—Varchar
 - (e) Numberofstudents—int

(iii) Insert the following data into the table :

SubjectCode	SubjectName	Date	Time	Numberofstudents
TPE602	DataSciene	1-3-2021	9:30-11:00	25
TCS101	Programming in C	2-3-2021	9:30-11:00	300
TMC105	Statistical Data Analysis	3-3-2021	2:30-4:00	200
TCS501	Software Engineering	4-3-2021	2:30-4:00	200
TPE601	Reservoir modelling and simulation	1-3-2021	2:30-4:00	29
XPE601	Career Skills	3-3-2021	2:30-4:00	32

5. (a) Table : Salesman

10 Marks (CO2)

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14
5007	Paul Adam	Rome	0.13
5003	LausonHen	San Jose	0.12

Table Name : Mast

PRO_ID	PRO_NAME	PRO_PRICE	PRO_COM
101	sulphur	3200	15
102	MS	450	16
103	HSD	250	14
104	Benzene	550	16
105	LPG	5000	11
106	NAPHTHA	900	12
107	PROPLENE	800	12
108	BITUMEN	2600	13
109	ATF	350	13
110	HEXANE	250	12

Table Name : Order

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.5	2012-10-05	3005	5002
70009	270.65	2012-09-10	3001	5005
70002	65.26	2012-10-05	3002	5001
70004	110.5	2012-08-17	3009	5003
70007	948.5	2012-09-10	3005	5002
70005	2400.6	2012-07-27	3007	5001
70008	5760	2012-09-10	3002	5001
70010	1983.43	2012-10-10	3004	5006
70003	2480.4	2012-10-10	3009	5003
70012	250.45	2012-06-27	3008	5002
70011	75.29	2012-08-17	3003	5007
70013	3045.6	2012-04-25	3002	5001

Considering above table write query for the following :

- Write a query to display the columns in a specific order like order date, salesman id, order number and purchase amount from for all the orders.
- Write a SQL query to find the name and price of the cheapest item(s) from the table.

P. T. O.

- (iii) Write a SQL statement to display names and city of salesman, who belongs to the city of Paris.
- (iv) Write the SQL statement to display the total amount of the sale done by James Hoog.

OR

(b) Database name-Midterm

10 Marks (CO2)

Table name—midtermschedule(column names: SubjectCode, SubjectName, Date, Time, Numberofstudents)

Write the SQL query for the following assuming the above database :

- (i) Print the subject name and subject of all subjects have midterm exam.
- (ii) Print the name of the subject where number of students appearing in the exam are less than 100.
- (iii) Print the table in sorted order according to the subject code.
- (iv) Print the details of the exam of those subjects for which subject code starts with TPE.

Roll No.

TPE-603

B. TECH. (PETROLEUM ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

NATURAL GAS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the quality lines in phase diagram. Draw the PT and PV phase diagram for single component system showing the different regions.

10 Marks (CO1, CO2)

OR

- (b) What do you mean by retrograde condensation ? Explain with the help of PT diagram.

10 Marks (CO1, CO2)

2. (a) What are the functions of fracturing fluids ? How fracturing fluid is made ? Discuss all the materials being used in making them.

10 Marks (CO2)

P. T. O.

OR

- (b) Calculate the gas specific gravity of natural gas in the reservoir having the following composition : 10 Marks (CO2)

Composition	Mole Fractions
C ₁	0.775
C ₂	0.083
C ₃	0.021
CO ₂	0.030
H ₂ S	0.020

The reservoir is having at initial condition of $P_i = 3500$ psia and 185°F . By calculating Sp. Gr. of the gas also calculate T_{pc} , P_{pc} , T_{pr} , P_{pr} by using Standing and Katz correlations (take compressibility factor $Z = 0.87$ for the data given) and hence the density of natural gas.

3. (a) Discuss the stability regions of methane hydrate and carbon dioxide hydrate on the PT curve by drawing the CH₄ hydrate equilibrium curve and CO₂ hydrate equilibrium curve. 10 Marks (CO1)

OR

- (b) What is natural gas ? Discuss all the properties of Natural gas with mathematical expressions. 10 Marks (CO1)

(3)

4. (a) What do you mean by proppant ? What are the functions of proppant and discuss the types of proppant. 10 Marks (CO1, CO2)

OR

- (b) With the help of diagram explain the extraction of natural gas by the enhanced (CBM) techniques from CBM ? Write down the usage and advantages of CBM. 10 Marks (CO1, CO2)
5. (a) Describe the steps involved in the extraction of shale gas. Why is hydraulic fracturing needed in the production of shale gas ?

10 Marks (CO1, CO2)

OR

- (b) Explain hydraulic fracturing. Illustrate the different theory and techniques for the hydraulic fracturing. 10 Marks (CO1, CO2)

Roll No.

TPE-604

B. TECH. (PETROLEUM ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

PETROLEUM DRILLING ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is directional drilling ? List and describe the four applications of directional drilling. 10 Marks (CO1)

OR

- (b) What data are required for directional well planning ? 10 Marks (CO1)

2. (a) Explain different types of well profile with the help of a diagram in directional drilling. 10 Marks (CO2)

OR

- (b) It is desired to drill under a lake to the location designated for Well 2. For this well, a build-and-hold trajectory will be used. Horizontal departure to the target is 2,655 ft at a TVD of 9,650 ft. The

P. T. O.

(2)

recommended rate of build is 2.00/100 ft. The kickoff depth is 1,600 ft. Determine (i) the radius of curvature, R_1 ; (ii) the maximum inclination angle, (iii) the measured depth to the end of the build. 10 Marks (CO2)

3. (a) State advantage and disadvantages of a standard removable whipstock as a deflection tool. 10 Marks (CO1)

OR

- (b) What is meant by 'multi-lobe' PDM's ? Explain. 10 Marks (CO1)

4. (a) Explain what is meant by reactive torque of downhole motors and why it causes problems for directional drillers. 10 Marks (CO1)

OR

- (b) When the drill pipe must be run into the hole while the annular preventers are closed, what is the difference between stripping and snubbing the pipe ? 10 Marks (CO1)

5. (a) Describe the different types of horizontal well, its advantage and disadvantage. 10 Marks (CO3)

OR

- (b) Design a well with two build up sections and a tangent section to land at the target at 90 degrees given : 10 Marks (CO3)

BUR first section = 5.5 deg

BUR second section = 10 deg

Target = 10000 ft TVD

Tangent angle = 50 deg

Tangent length = 500 ft

Calculate first and second KOP depths.

Roll No.

TPE-605

B. TECH. (PETROLEUM ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

PETROLEUM PRODUCTION ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All question compulsory.

(ii) Answer any one sub-question between (a) and (b) in each main question.

(iii) Total marks for each main question are ten.

1. (a) Differentiate between SPAR platform and semi-sub platform.

10 Marks (CO1, CO2)

OR

(b) Explain subsea production system and write all the equipments used for Subsea production.

10 Marks (CO1, CO2)

2. (a) Explain what is well activation and how compressed air can be used for well activation.

10 Marks (CO2)

P. T. O.

(2)

OR

- (b) Why perforation is done and what are shaped charge perforation operations ? How corrosion protection is done in offshore platforms and how oil is transported from offshore oil and gas facility ?

10 Marks (CO2)

3. (a) Differentiate between Concrete gravity structures and tension leg platforms and also explain compliant towers. 10 Marks (CO1, CO2)

OR

- (b) Explain FPSO and also write main parts of a platform.

10 Marks (CO1, CO2)

4. (a) Explain different types of loads on offshore structures. Explain any *four*. 10 Marks (CO1, CO2)

OR

- (b) Write a short note on safety systems in production facilities.

10 Marks (CO1, CO2)

5. (a) Explain what offshore platforms are and in how many types they can be categorized, name them. 10 Marks (CO1, CO2)

OR

- (b) What is production riser ? Explain steel catenary riser and well completion. 10 Marks (CO1, CO2)

50

Roll No.

TPE-611

B. TECH. (PE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022

ALTERNATE ENERGY RESOURCES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) List various non-conventional energy resources. Give their availability, relative merits, demerits and their classification.

10 Marks (CO1/CO2/CO3)

OR

- (b) What is the status of non-conventional energy sources in India and what are their future prospects ?

10 Marks (CO1/CO2/CO3)

2. (a) Distinguish between renewable and non-renewable energy sources.

10 Marks (CO3/CO4)

OR

- (b) Define the term "Solar Insolation". State the parameters of solar insolation for a given flat surface.

10 Marks (CO3/CO4)

P. T. O.

(2)

3. (a) Explain the importance of various non-conventional energy sources in the context of global warming. 10 Marks (CO1/CO4)

OR

- (b) Classify solar collectors. State advantages, disadvantages and working with neat diagram of "Flat Plate collectors". 10 Marks (CO1/CO4)
4. (a) What is "Photovoltaic cell" ? Explain briefly with a neat diagram a "silicon photovoltaic cell". 10 Marks (CO4)

OR

- (b) Define solar irradiance, solar constant, extraterrestrial and terrestrial radiations. What is the standard value of solar constant ? 10 Marks (CO4)
5. (a) Describe percentage wise distribution of various components (solar spectrum) in extraterrestrial radiation. 10 Marks (CO4)

OR

- (b) What is a Solar Still ? Draw its diagram and explain its working in detail. 10 Marks (CO4)